JAYSON QUINDAO

SENIOR SYSTEMS & DISTRIBUTED INFRASTRUCTURE ENGINEER • RUST / HIGH-CONCURRENCY

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PROFESSIONAL SUMMARY

Systems & Distributed Infrastructure Engineer with 6+ years building low-latency runtimes, database engine internals, and kernel/OS-level telemetry in Rust and C/C++. Active upstream contributor to Qdrant Vector Database (PR #11432, PR #11428) resolving index overflow and facet regressions, and Turso's distributed SQLite engine (libSQL PR #2309) resolving C-FFI memory leaks via RAII reclamation. Architect of open-source Rust telemetry architecture with Win32 process enumeration and HMAC-SHA256 event streams. Proven record deploying enterprise telemetry across national ATM networks at NCR Atleos and high-throughput biometric pipelines at Mentalix.

TECHNICAL COMPETENCIES

• **Languages & Systems:** Rust (Edition 2021, Cargo, Tokio, Serde, FFI / C-ABI, Unsafe Boundaries, RAII, Concurrency), C/C++, Go, Python (FastAPI), TypeScript, Bash/Shell, SQL (PostgreSQL, SQLite WAL).

• **Distributed Databases & Networking:** Qdrant Vector Engine, Distributed SQLite (libSQL/Turso), WebSockets, gRPC, TCP/UDP Socket Programming, Event-Driven Streaming, Zero-Copy Serialization (Protobuf, FlatBuffers).

• **Security & Telemetry:** Cryptographic HMAC-SHA256 Payload Signing, OS-Level Process Inspection (Win32 K32EnumProcesses, OpenProcess, POSIX), AES-256 Encryption, RBAC, CJIS/NIST Compliance.

• **Cloud & Tooling:** Linux Kernel/POSIX Internals, Docker, GitHub Actions CI/CD, AWS/Cloudflare, Git, Vitest, Deterministic Test Harnesses.

UPSTREAM OPEN-SOURCE DATABASE CONTRIBUTIONS

QDRANT — Vector Search Engine Upstream Contributor github.com/qdrant/qdrant/pull/11428

Stack: Rust, Vector Index Internals, Edge Shard, Exact Facet Parity, Integration Testing

• **Exact Facet Zero-Count Omission (PR #11428):** Resolved an open regression in Qdrant's embedded Edge shard (#11409) where exact facet counts listed moved or deleted values with count: 0, violating parity with approximate queries.

• **Guarded Collection Filtering:** Wrapped exact count collection in live positive-count verification and authored regression test exact_facet_after_move() covering keyword indexes after point mutations.

TURSO / LIBSQL — Distributed SQLite Engine Contributor github.com/tursodatabase/libsql/pull/2309

Stack: Rust, C-FFI, SQLite Engine Internals, Arc/Box Memory Management, Integration Testing

• **C-FFI Memory Leak Remediation (PR #2309):** Diagnosed and resolved open memory leak in SQLite's update hook callback mechanism (#2295) where raw pointer user context persisted across connection termination.

• **RAII Safe Pointer Reclamation:** Implemented explicit context reclamation via Box::from_raw during connection teardown, preventing memory leaks and maintaining strict C-ABI stability across 28 automated test suites.

PROFESSIONAL EXPERIENCE

GG LOOP LLC — Founder & Principal Systems Architect Nov 2023 – Present | Dallas, TX

Stack: Rust, Cargo, Tokio, Win32 API, C-ABI FFI, Serde, WebSockets • github.com/djjrip/anti-cheat-sdk

• **High-Performance Rust Telemetry SDK:** Built zero-overhead telemetry SDK utilizing OS-level handles and process enumeration to monitor system execution in real-time with sub-2ms latency via Tokio and crossbeam channels.

• **Memory Safety & Native FFI:** Structured unsafe Rust boundaries wrapping low-level Win32/POSIX APIs (K32EnumProcesses, OpenProcess), implementing RAII handle cleanup that eliminated memory leaks and guaranteed zero crash vulnerability.

• **Cryptographic Packet Signing:** Engineered real-time cryptographic payload integrity checks generating HMAC-SHA256 signatures for telemetry packets, preventing replay attacks and stream desynchronization.

NCR ATLEOS — Systems & Data Automation Engineer Aug 2021 – Nov 2023 | Frisco, TX

Stack: Python, SQL (PostgreSQL, T-SQL), Distributed Banking Hardware Telemetry, Network Monitoring

• **Distributed Hardware Telemetry:** Monitored and reconciled multi-million dollar daily transaction streams across national ATM hardware endpoints and banking network gateways, maintaining 99.9%+ availability.

• **40% Anomaly Turnaround Reduction:** Engineered automated anomaly-detection scripts that slashed discrepancy investigation cycle times by 40% across multi-tenant banking data feeds.

MENTALIX, INC. — Data Security & Systems Pipeline Engineer Sep 2020 – Sep 2021 | Plano, TX

Stack: C++, Python, Hardware-Software Interfacing, AES-256, CJIS/NIST Compliance

• **Hardware-Software Interfacing:** Developed high-volume biometric data capture pipelines interfacing directly with physical hardware scanning devices and peripheral controllers for enterprise and federal clients.

• **Embedded Security & Compliance:** Enforced hardware-level AES-256 encryption, strict RBAC, and CJIS standards on streaming biometric packets, speeding throughput by 25% while passing 100% of security audits.

EDUCATION & CITIZENSHIP

UNIVERSITY OF TEXAS AT ARLINGTON — Bachelor of Arts (Broadcasting / Digital Communications) Arlington, TX

Coursework in Digital Communications, Network Media Systems, and Telemetry. 73 credits transferred from TCC.

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